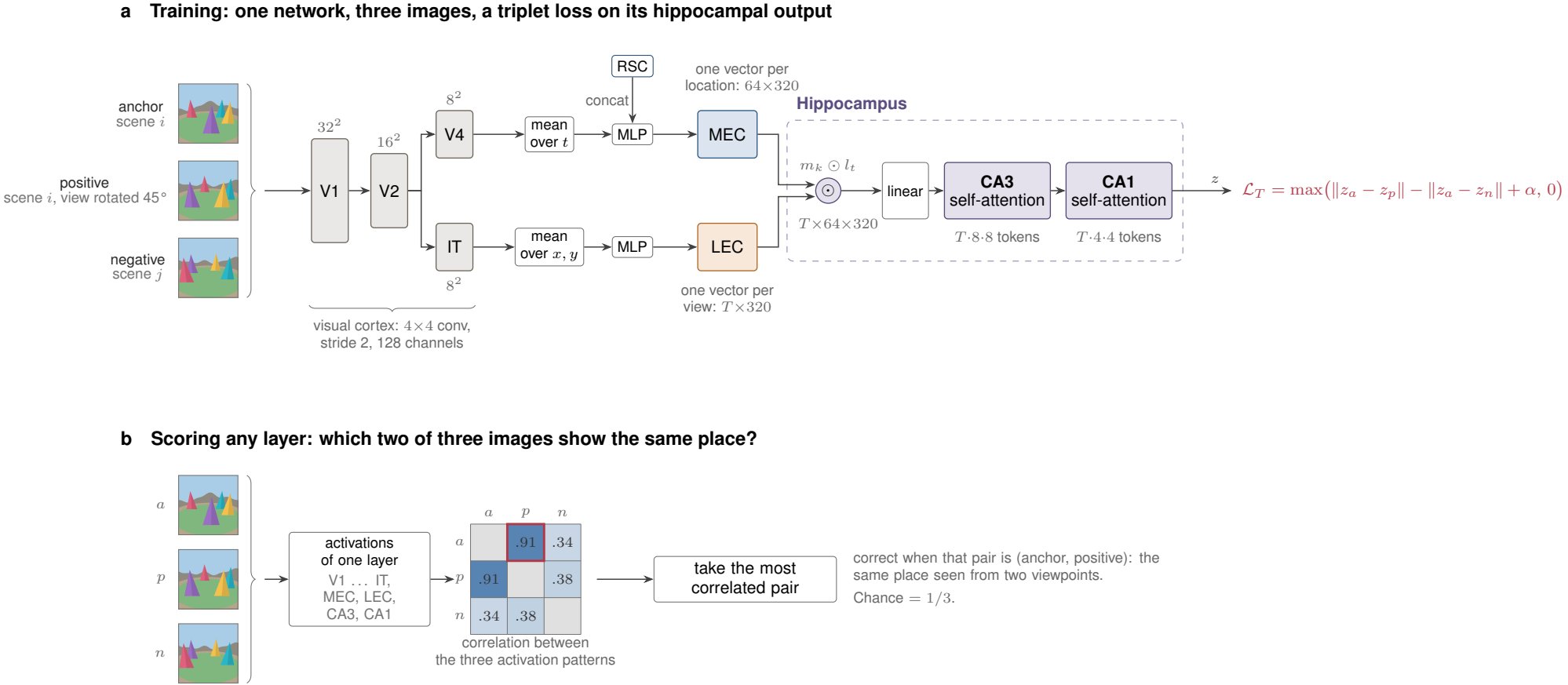

a Training: one network, three images, a triplet loss on its hippocampal output
anchor
scene i
positive
scene i, view rotated 45°
negative
scene j
32^2
V1
16^2
V2
8^2
V4
IT
8^2
visual cortex: 4×4 conv, stride 2, 128 channels
mean over t
mean over x, y
RSC
concat
MLP
MLP
one vector per location: 64×320
MEC
LEC
one vector per view: $T\times320$
Hippocampus
$m_k \odot l_t$
$T\times64\times320$
linear
CA3 self-attention
$T\cdot8\cdot8$ tokens
CA1 self-attention
$T\cdot4\cdot4$ tokens
z
$\mathcal{L}_T = \max(\|z_a - z_p\| - \|z_a - z_n\| + \alpha, 0)$
b Scoring any layer: which two of three images show the same place?
a
p
n
activations of one layer
V1 … IT, MEC, LEC, CA3, CA1
a p n
a .91 .34
p .91 .38
n .34 .38
correlation between the three activation patterns
take the most correlated pair
correct when that pair is (anchor, positive): the same place seen from two viewpoints.
Chance $= 1/3$.